

3

Isoparametric Elements and Elementary Programming Concepts

3.1 PRELIMINARY CONCEPTS

We wish to define the shape functions in such a way that, as the finite element mesh is refined, the approximate Galerkin solution converges to the exact solution. The following question arises: What conditions must the shape functions satisfy so that this property is guaranteed? We shall be content, for the time being, to state sufficient conditions for convergence. These conditions are possessed by the most prevalent and important finite element shape functions. However, we note that convergent elements can be constructed from shape functions which do not satisfy all these requirements. Nevertheless these conditions may be considered basic in that they provide the simplest criteria to ensure convergence for a wide class of problems.

The basic convergence requirements are that the shape functions be

- C1. smooth (i.e., at least C^1) on each element interior, Ω^e ;
- C2. continuous across each element boundary Γ^e ; and
- C3. complete.

Remarks

1. Conditions C1 and C2 guarantee that first derivatives of the shape functions have, at worst, finite jumps across the element interfaces; see Fig. 3.1.1. This ensures that all integrals necessary for the computation of element arrays (see Sec. 2.11) are well defined, since at most first derivatives appear in the integrands. If we permit finite discontinuities in the shape functions on element boundaries, the derivatives possess delta functions (cf. Sec. 1.10) and we are unable to make sense out of the squares of these quantities that would appear in the stiffness integrands.

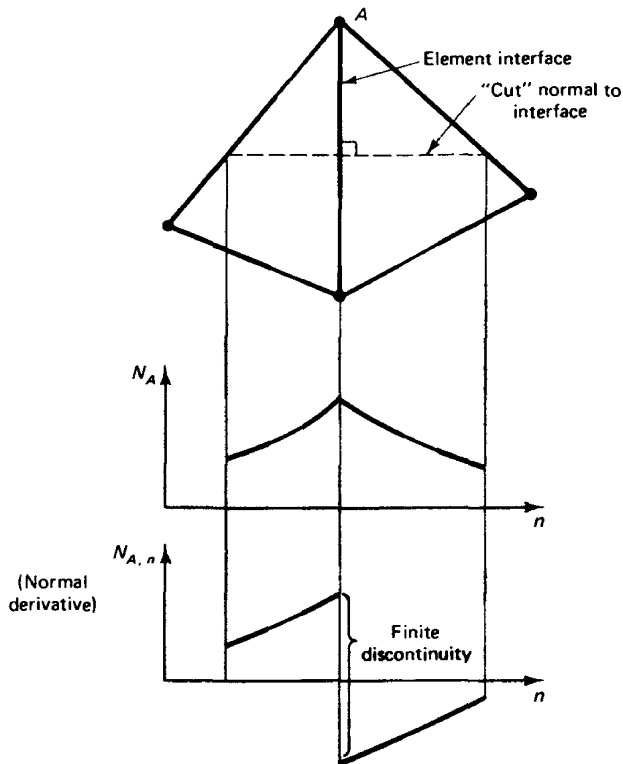


Figure 3.1.1 Example of a shape function that satisfies conditions C1 and C2.

2. Shape functions that satisfy C1 and C2 are of class $C^0(\bar{\Omega})$. Finite elements constructed from $C^0(\bar{\Omega})$ shape functions are often referred to as C^0 -elements.

3. Theories such as Bernoulli-Euler beam theory, which require higher-order derivatives in the stiffness integrands, necessitate strengthening condition C1 to C^2 -continuity on Ω^e and condition C2 to C^1 -continuity across Γ^e . Shape functions of this type are of class $C^1(\bar{\Omega})$ and lead to so-called C^1 -elements. Examples of $C^1(\bar{\Omega})$ shape functions are the Hermite cubics discussed in Sec. 1.16.

4. If the stiffness integrands involve derivatives of order m , Condition C1 should be strengthened to C^m -continuity on Ω^e and Condition C2 should be strengthened to C^{m-1} -continuity across Γ^e . Finite elements that satisfy this property are called *conforming*, or *compatible*. (The use of elements that violate this property, *non-conforming* or *incompatible elements*, is, however, common. We say more about this later.)

5. It may be noted that C2 is equivalent to requiring that each function $u^h \in \mathcal{S}^h$ be continuous across Γ^e . In specific cases it is often easier to establish this fact directly rather than prove it for individual shape functions.

Completeness

To illustrate the completeness requirement, let us take the case of heat conduction in which the e th element interpolation function may be written

$$u^h = \sum_{a=1}^{n_{eq}} N_a d_a^e \quad (3.1.1)$$

where $d_a^e = u^h(\mathbf{x}_a^e)$. Let $n_{sd} = 3$. In this case, the shape functions are said to be **complete** if

$$d_a^e = c_0 + c_1 x_a^e + c_2 y_a^e + c_3 z_a^e \quad (3.1.2)$$

implies that

$$u^h(\mathbf{x}) = c_0 + c_1 x + c_2 y + c_3 z \quad (3.1.3)$$

where $c_0, \dots, c_3$ are arbitrary constants. In two dimensions, we omit the z -terms in (3.1.2) and (3.1.3). In words, *completeness requires that the element interpolation function is capable of exactly representing an arbitrary linear polynomial when the nodal degrees of freedom are assigned values in accordance with it.*

Completeness is a plausible requirement as the following argument indicates: As the finite element mesh is further and further refined, the exact solution and its derivatives approach constant values over each element domain. To ensure that these constant values are representable, the shape functions must contain all constant and linear monomials. This argument was originally given in [1] and has subsequently been proved to be the key mathematical idea for proving convergence theorems for finite element approximations (e.g., see [2]).

For elasticity, the requirement is essentially the same. In this case we may write

$$u_i^h = \sum_{a=1}^{n_{eq}} N_a d_{ia}^e \quad (3.1.4)$$

where $d_{ia}^e = u_i^h(\mathbf{x}_a)$. The N_a are complete if

$$d_{ia}^e = c_0 + c_1 x_a^e + c_2 y_a^e + c_3 z_a^e \quad (3.1.5)$$

implies

$$u_i^h(\mathbf{x}) = c_0 + c_1 x + c_2 y + c_3 z \quad (3.1.6)$$

Remarks

6. In elasticity, the presence of all monomials through linear terms means that an element may exactly represent all rigid motions (i.e., rigid translations and rotations) and constant strain states. We thus often speak of completeness in terms of the ability to represent rigid motions and constant strains (e.g., see [3], p. 33).

7. For theories involving m th derivatives in the stiffness integrands, completeness must be strengthened to m th-order polynomials.

Example 1

The piecewise linear finite element space introduced in Chapter 1 satisfies the convergence conditions C1–C3. C1 and C2 follow from the definition (see Sec. 1.8). For C3 we proceed as follows: Assume $d_a^e = c_0 + c_1 x_a^e$ and substitute in $u^h = \sum_{a=1}^2 N_a d_a^e$, viz.,

$$\begin{aligned}
u^h &= \sum_{a=1}^2 N_a d_a^e \\
&= \sum_{a=1}^2 N_a (c_0 + c_1 x_a^e) \\
&= \left(\sum_{a=1}^2 N_a \right) c_0 + \left(\sum_{a=1}^2 N_a x_a^e \right) c_1
\end{aligned} \tag{3.1.7}$$

Thus to establish completeness we must show that

$$\sum_{a=1}^2 N_a = 1 \tag{3.1.8}$$

and

$$\sum_{a=1}^2 N_a x_a^e = x \tag{3.1.9}$$

By (1.12.5), $N_a(\xi) = (1 + (-1)^a \xi)/2$, from which (3.1.8) follows. Equation (3.1.9) was established previously (see (1.12.6)).

3.2 BILINEAR QUADRILATERAL ELEMENT

This element is attributed to Taig [4]. The rectangular version was proposed earlier by Argyris in [5].

The domain of a straight-edged quadrilateral element is defined by the locations of its four nodal points x_a^e , $a = 1, \dots, 4$ in the $\mathbb{R}^2$ -plane. We assume the nodal points are labeled in ascending order corresponding to the counterclockwise direction (see Fig. 3.2.1). We seek a change of coordinates which maps the given quadrilateral into the biunit square, as depicted in Fig. 3.2.1. The biunit square is sometimes called the *parent domain*. The coordinates of a point

$$\xi = \begin{Bmatrix} \xi \\ \eta \end{Bmatrix} \tag{3.2.1}$$

in the biunit square are to be related to the coordinates of a point

$$x = \begin{Bmatrix} x \\ y \end{Bmatrix} \tag{3.2.2}$$

in Ω^e by mappings of the form

$$x(\xi, \eta) = \sum_{a=1}^4 N_a(\xi, \eta) x_a^e \tag{3.2.3}$$

$$y(\xi, \eta) = \sum_{a=1}^4 N_a(\xi, \eta) y_a^e \tag{3.2.4}$$

ξ and η are sometimes called the *natural coordinates*. A more succinct representation of the above formulas is

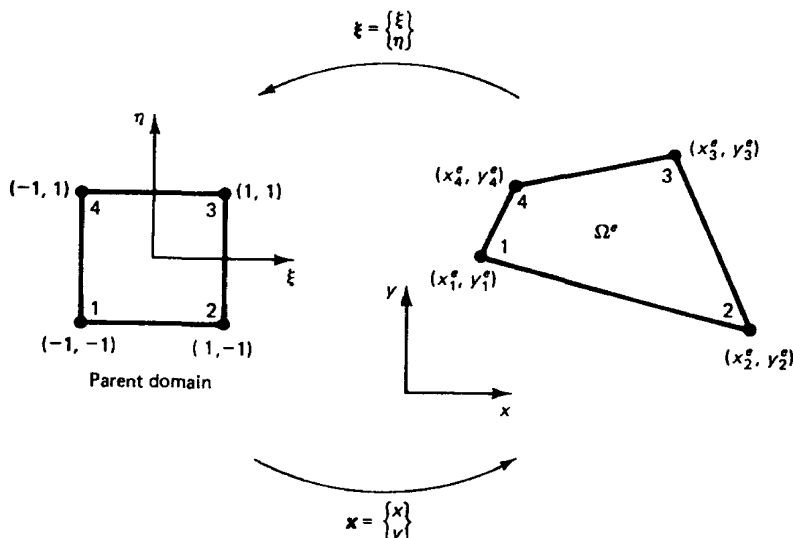


Figure 3.2.1 Bilinear quadrilateral element domain and local node ordering.

$$\mathbf{x}(\xi) = \sum_{a=1}^4 N_a(\xi) \mathbf{x}_a^e \quad (3.2.5)$$

We obtain the functions N_a by first *assuming* the “bilinear” expansions

$$x(\xi, \eta) = \alpha_0 + \alpha_1 \xi + \alpha_2 \eta + \alpha_3 \xi \eta \quad (3.2.6)$$

$$y(\xi, \eta) = \beta_0 + \beta_1 \xi + \beta_2 \eta + \beta_3 \xi \eta \quad (3.2.7)$$

where the α 's and β 's are parameters to be determined; and second, by stipulating that (3.2.6) and (3.2.7) satisfy the (respective) conditions

$$x(\xi_a, \eta_a) = x_a^e \quad (3.2.8)$$

$$y(\xi_a, \eta_a) = y_a^e \quad (3.2.9)$$

where ξ_a and η_a are defined in Table 3.2.1.

TABLE 3.2.1 Nodal coordinates in ξ -space

a	ξ_a	η_a
1	-1	-1
2	1	-1
3	1	1
4	-1	1

Conditions (3.2.8) and (3.2.9) impose restrictions on the functions N_a ; namely,

$$N_a(\xi_b) = \delta_{ab} \quad (3.2.10)^1$$

To see this we can combine (3.2.3) and (3.2.8), viz.,

$$x_b^e = x(\xi_b, \eta_b) = \sum_{a=1}^4 N_a(\xi_b, \eta_b) x_a^e \quad (3.2.11)$$

which holds only if $N_a(\xi_b, \eta_b) = \delta_{ab}$. This same restriction may be deduced by combining (3.2.4) and (3.2.9). Equations (3.2.6) through (3.2.9) lead to the following matrix equations:

$$\begin{Bmatrix} x_1^e \\ x_2^e \\ x_3^e \\ x_4^e \end{Bmatrix} = \begin{bmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \begin{Bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{Bmatrix} \quad (3.2.12)$$

$$\begin{Bmatrix} y_1^e \\ y_2^e \\ y_3^e \\ y_4^e \end{Bmatrix} = \begin{bmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \begin{Bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{Bmatrix} \quad (3.2.13)$$

In each case the coefficient matrix is the same. Solving these for the α 's and β 's, substituting the results into (3.2.6) and (3.2.7), and comparing with (3.2.3) and (3.2.4) reveals that

$$N_a(\xi) = N_a(\xi, \eta) = \frac{1}{4}(1 + \xi_a \xi)(1 + \eta_a \eta)$$

(3.2.14)

Note that this is precisely the product of one-dimensional shape functions derived previously (see (1.12.5)).

Exercise 1. Verify (3.2.14).

Observe that coordinate lines in ξ -space (i.e., lines such that $\xi = \text{constant}$ or $\eta = \text{constant}$) are mapped into straight lines in x -space. In particular, boundary lines are mapped into boundary lines (see Fig. 3.2.2).

Now we shall *assume* that the element functions are given by similar expansions in terms of the shape functions.

Heat conduction

$$u^h(\xi) = \sum_{a=1}^4 N_a(\xi) d_a^e \quad (3.2.15)$$

¹Equation (3.2.10) is sometimes referred to as the *interpolation property* since it implies that the assumed expansions (e.g., (3.2.6) and (3.2.7)) interpolate the nodal data (e.g., (3.2.8) and (3.2.9)).

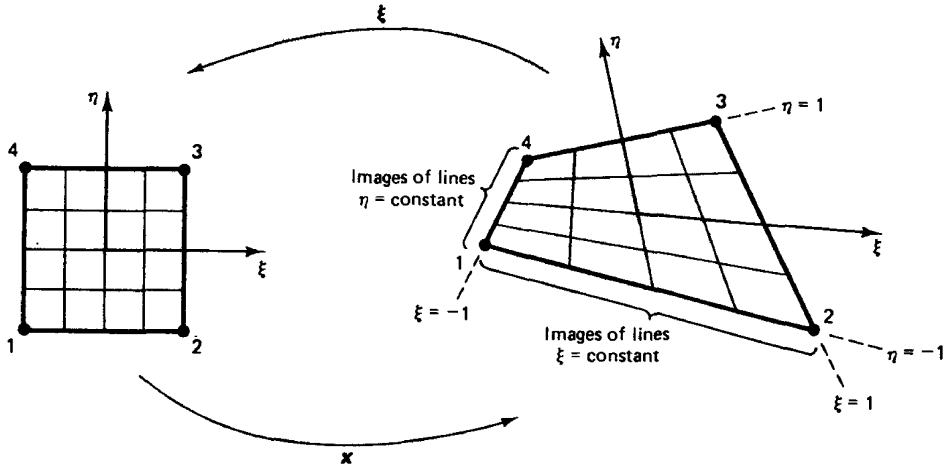


Figure 3.2.2

Elastostatics

$$u_i^h(\xi) = \sum_{a=1}^4 N_a(\xi) d_{ia}^e \quad (3.2.16)$$

Henceforce, it will be sufficient to consider only (3.2.15). The case of elastostatics may be deduced by viewing (3.2.15) as any one of the n_{sd} component functions of (3.2.16).

(C1) Smoothness on Ω^e . It can be shown that N_a is a smooth function of x and y , if all interior angles formed by two adjacent edges are less than 180° ; see Fig. 3.2.3. (Note that N_a is always a smooth function of ξ and η ; see (3.2.14). When N_a is viewed as a function of x and y (i.e., $N_a(\xi(x, y), \eta(x, y))$), to ascertain smoothness we must consider the inverse functions $\xi(x, y)$ and $\eta(x, y)$, which may not be well defined if the element is too distorted. Figure 3.2.3 illustrates when this occurs for the four-node element. This issue is discussed from a more general perspective in the following section.)

(C2) Continuity across Γ^e . Let us examine a typical shape function N_a . For the moment, assume $a = 1$ or 2 . The behavior of N_a along the edge connecting nodes 1 and 2 may be determined by substituting $\eta = -1$ into (3.2.14). The result is

$$N_a(\xi, -1) = \frac{1 + \xi_a \xi}{2}, \quad a = 1, 2 \quad (3.2.17)$$

From Table 3.2.1, $\xi_1 = -1$ and $\xi_2 = +1$; thus the right-hand side of (3.2.17) is the familiar linear shape function of the one-dimensional theory (see Fig. 3.2.4). By virtue of the fact that (3.2.17) is typical of all four edges and by (3.2.10), we conclude

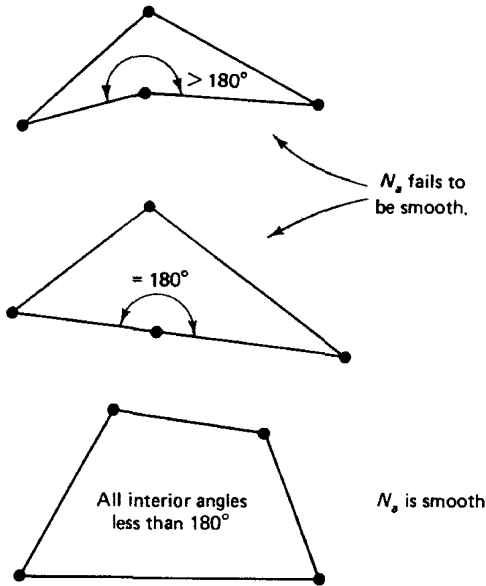


Figure 3.2.3 Smoothness criterion for bilinear shape functions.

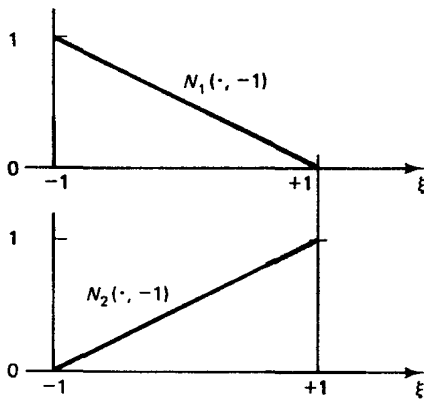


Figure 3.2.4 Bilinear shape functions along edge connecting nodes 1 and 2.

that N_a appears as in Fig. 3.2.5. N_a is said to be a *hyperbolic paraboloid*. Similar considerations for adjacent elements indicate that N_A is continuous and appears as a “tent” whose “pole” emanates from node A ; see Fig. 3.2.6 for a typical situation at an internal node. Consequently, $\mathcal{J}^h$ consists of continuous functions as each member is a linear combination of the N_A ’s (cf. (2.4.3)–(2.4.5)). It is instructive to argue this fact from a slightly different point of view.

Consider the behavior of (3.2.15) along the boundary segment connecting x_1^* to x_2^* . By (3.2.17), this may be written

$$u^h(\xi, -1) = \sum_{a=1}^2 \frac{1}{2}(1 + \xi_a \xi) d_a^e \quad (3.2.18)$$

There are two important observations to be made about this result:

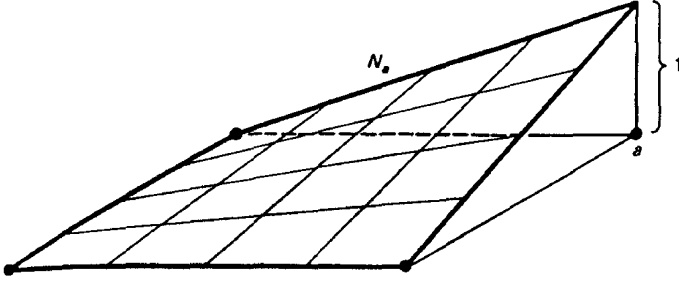


Figure 3.2.5 Bilinear shape function.

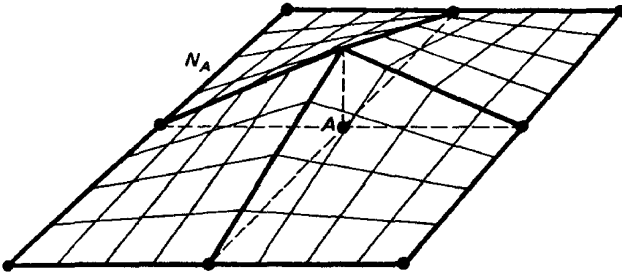
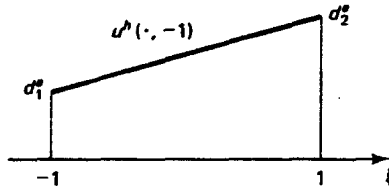


Figure 3.2.6 Global shape function constructed from element bilinear shape functions.

1. The behavior of u^h along $[x_1^e, x_2^e]$ is determined solely by the nodal values of u^h at the nodes x_1^e and x_2^e .
2. The variation of u^h is linear in the natural coordinate ξ (see Fig. 3.2.7).


 Figure 3.2.7 Graph of u^h along boundary segment joining nodes x_1^e and x_2^e .

The same conclusion can be drawn for any other element of this type whose boundary contains $[x_1^e, x_2^e]$. Thus we see that continuity of u^h is automatic across element interfaces.

(C3) Completeness

$$\begin{aligned}
 u^h &= \sum_{a=1}^{n_{eq}} N_a d_a^e \\
 &= \sum_{a=1}^{n_{eq}} N_a (c_0 + c_1 x_a^e + c_2 y_a^e) \\
 &= \left(\sum_{a=1}^{n_{eq}} N_a \right) c_0 + \underbrace{\left(\sum_{a=1}^{n_{eq}} N_a x_a^e \right)}_{= x \text{ by (3.2.3)}} c_1 + \underbrace{\left(\sum_{a=1}^{n_{eq}} N_a y_a^e \right)}_{= y \text{ by (3.2.4)}} c_2
 \end{aligned}$$

Thus it remains only to prove that the shape functions sum to value 1 at each point. This may be done by explicit calculation, viz.,

$$\begin{aligned} \sum_{a=1}^{n_{en}} N_a(\xi, \eta) &= \frac{1}{4}(1 - \xi)(1 - \eta) + \frac{1}{4}(1 + \xi)(1 - \eta) \\ &\quad + \frac{1}{4}(1 + \xi)(1 + \eta) + \frac{1}{4}(1 - \xi)(1 + \eta), \quad (\text{by (3.2.14)}) \\ &= 1 \end{aligned} \quad (3.2.20)$$

3.3 ISOPARAMETRIC ELEMENTS

The isoparametric concept is generally attributed to Taig [6] and Irons [7].

Let $\square$ denote a parent domain in ξ -space (e.g., the closed, biunit square in $\mathbb{R}^2$).

Definition 1. Let $x: \square \rightarrow \bar{\Omega}^e$ be of the form

$$x(\xi) = \sum_{a=1}^{n_{en}} N_a(\xi) x_a^e \quad (3.3.1)$$

If the element interpolation function u^h can be written as

$$u^h(\xi) = \sum_{a=1}^{n_{en}} N_a(\xi) d_a^e \quad (3.3.2)$$

the element is said to be *isoparametric*.

The key point to observe in the definition is that the shape functions which define (3.3.1) also serve to define (3.3.2).

The bilinear quadrilateral, derived in the previous section, is an example of an isoparametric element.

We shall now discuss some fundamental mathematical properties of mappings. Based on this, we shall argue that convergence condition C1 is generally achieved for isoparametric elements. Furthermore, a criterion, which may be used in a computer program to determine whether or not individual elements satisfy C1, emanates from the discussion.

Convergence Condition C1

Definition 2. A mapping $x: \square \rightarrow \bar{\Omega}^e \subset \mathbb{R}^{n_{sd}}$ is said to be *one-to-one* if for each pair of points $\xi^{(1)}, \xi^{(2)} \in \square$ such that $\xi^{(1)} \neq \xi^{(2)}$, then $x(\xi^{(1)}) \neq x(\xi^{(2)})$.

In words, this statement means that two different points of $\square$ do not get mapped into the same point in $\bar{\Omega}^e$.

Definition 3. $x: \square \rightarrow \bar{\Omega}^e$ is said to be *onto* if $\bar{\Omega}^e = x(\square)$ (i.e., each point in $\bar{\Omega}^e$ is the image of a point in $\square$ under the mapping x).

Definition 4. Let $x : \square \rightarrow \bar{\Omega}^e$ be a differentiable mapping. The determinant of the derivative, denoted by $j = \det(\partial x / \partial \xi)$, is called the **Jacobian determinant**.

The Jacobian determinants in two and three dimensions, respectively, are given explicitly by

$$j = \det \begin{bmatrix} x_{,\xi} & x_{,\eta} \\ y_{,\xi} & y_{,\eta} \end{bmatrix} \quad (3.3.3)$$

$$j = \det \begin{bmatrix} x_{,\xi} & x_{,\eta} & x_{,\zeta} \\ y_{,\xi} & y_{,\eta} & y_{,\zeta} \\ z_{,\xi} & z_{,\eta} & z_{,\zeta} \end{bmatrix} \quad (3.3.4)$$

Remark

1. It is a consequence of the inverse function theorem (e.g., see [8]) that if x is

- i. one-to-one;
- ii. onto;
- iii. C^k , $k \geq 1$; and if
- iv. $j(\xi) > 0$ for all $\xi \in \square$;

then the inverse mapping $\xi = x^{-1} : \bar{\Omega}^e \rightarrow \square$ exists and is C^k .

Proposition 1. Let the mapping defined by (3.3.1) satisfy (i) through (iv). Then the smoothness requirement (C1) is satisfied.

Proof. By virtue of the hypotheses, $N_a = N_a(\xi)$ is also a C^1 function. By Remark 1, $\xi = \xi(x)$ is also C^1 . Thus $N_a(x) = N_a(\xi(x))$ is a C^1 function of x . (This last fact may be proved with the aid of the chain rule.) ■

Remarks

2. In practice, the mappings $x : \square \rightarrow \bar{\Omega}^e$ usually satisfy (i) through (iv). However, there is one exception of practical importance. It is concerned with the technique of element "degeneration," in which nodes are coalesced. The simplest example of this procedure, in which two nodes of the bilinear quadrilateral are coalesced to form a triangle, is presented in the next section. Further examples are presented in subsequent sections. When degeneration is performed, the Jacobian determinant vanishes at certain nodal points within the element. Away from these points it is positive, and the mapping $\xi = \xi(x)$ remains smooth (i.e., C1 is satisfied). For reasons that will be apparent later on, it is not usually required to calculate derivatives at these points.

3. In actual computations the sign of the Jacobian determinant is monitored at special points to ensure condition (iv) is satisfied. If a zero or negative value is encountered, computations are terminated. Generally this is an indication of an input-data error or an overly distorted element.

Convergence Condition (C3)

Proposition 2. If $\sum_{a=1}^{n_{en}} N_a = 1$, then completeness condition C3 is satisfied for isoparametric elements.

Proof. (We shall prove the assertion for the three-dimensional case.)

$$\begin{aligned}
 u^h &= \sum_{a=1}^{n_{en}} N_a d_a^e && \text{(by (3.3.2))} \\
 &= \sum_{a=1}^{n_{en}} N_a (c_0 + c_1 x_a^e + c_2 y_a^e + c_3 z_a^e) \\
 &= c_0 \left(\sum_{a=1}^{n_{en}} N_a \right) + c_1 \left(\sum_{a=1}^{n_{en}} N_a x_a^e \right) + c_2 \left(\sum_{a=1}^{n_{en}} N_a y_a^e \right) + c_3 \left(\sum_{a=1}^{n_{en}} N_a z_a^e \right) \\
 &= c_0 \left(\sum_{a=1}^{n_{en}} N_a \right) + c_1 x + c_2 y + c_3 z && \text{(by (3.3.1))}
 \end{aligned}$$

The condition that the shape functions sum to one, assumed in Proposition 2, is easily checked on a case-by-case basis. ■

The only remaining convergence condition is C2, the continuity requirement on Γ^* . This condition can be verified once the construction of the global shape functions from the element shape functions is explicated. It happens that if this procedure is done in the “obvious” way, continuity is achieved. In the sequel we shall consider this issue on a case-by-case basis.

Summary. The importance of the isoparametric concept is that the three basic convergence conditions are virtually automatic. In addition, isoparametric elements may be designed to take on convenient shapes for practical analysis, including curved boundaries, and lend themselves to concise computer implementation.

3.4 LINEAR TRIANGULAR ELEMENT; AN EXAMPLE OF “DEGENERATION”

The linear triangle was one of the first finite elements conceived (see Courant [9] and Turner et al. [10]) and is still widely used. In the structural mechanics literature it is often referred to as the constant stress/strain triangle, or simply the CST. We shall derive it from the bilinear quadrilateral by coalescing nodes 3 and 4; see Fig. 3.4.1. Specifically, we set $x_4^e = x_3^e$ in (3.2.5) and define new shape functions for the triangle N'_a , $a = 1, 2, 3$, as follows:

$$\begin{aligned}
 x &= \sum_{a=1}^4 N_a x_a^e \\
 &= N_1 x_1^e + N_2 x_2^e + (N_3 + N_4) x_3^e = \sum_{a=1}^3 N'_a x_a^e \quad (3.4.1)
 \end{aligned}$$